

PROMPT ENGINEERING

Complete Study Notes

GeeksforGeeks Article × ABTalks Handbook

Edition 2025 • 21 Chapters • 50 Templates

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1. What is Prompt Engineering?

Prompt engineering is the process of creating clear and effective prompts that guide AI models to generate accurate responses. It focuses on writing smart prompts for text-based AI (especially NLP) so the model produces the required output with minimal back-and-forth.

1.1 Why It Matters

- More specific prompts improve interpretability of task requirements.
- Specific prompts with detailed requirements produce outputs closer to the desired one.
- Better prompts train the model more effectively for future tasks.
- Saves time — reduces correction cycles and back-and-forth.
- Unlocks capabilities most users never discover.
- Essential for freelancing, consulting, product development, and enterprise AI.
- Applicable across writing, coding, analysis, design, education, and marketing.

1.2 What Are Prompts?

Prompts are short pieces of text that provide context and guidance to machine learning models. For NLP tasks they help generate relevant outputs by:

- Adding additional guidance for the model.
- Avoiding over-generalisation of the prompt.
- Keeping information concise — too much can confuse the model.
- Making the user intent and purpose clear so the model stays in relevant context.

1.3 How Prompt Engineering Works

Think of it like instructing a talented but inexperienced assistant. You craft the right instructions (prompts) to get the desired result from a Large Language Model (LLM). The process involves three phases:

Phase	Description
1. Craft the Prompt	Design a question, statement, or example — wording, phrasing, and context all guide the LLM's response
2. Understand the LLM	Choose the right technique: zero-shot (minimal instructions) or few-shot (examples provided).
3. Refine the Prompt	Trial-and-error iteration — tweak based on output until you get the desired response.

2. Applications of Prompt Engineering

Prompt engineering is used most heavily in text-based modelling and NLP. Key application areas:

Application	How Prompts Help
Language Translation	Include script, dialect, and contextual features for more accurate translations.
Q&A Chatbots	Add intent and context so the bot isn't confused when generating relevant answers.
Text Generation	Specify purpose, tone, vocabulary, and structure to shape the generated content.

3. The Prompt Quality Ladder

Level	Example Prompt	Why It Works / Fails
Weak	write about marketing	Too vague — produces a generic essay.
Okay	write 5 marketing tips for a small bakery	Better but lacks persona and format.
Good	Act as a marketing consultant. 5 low-budget Instagram ideas. Bullets.	Persona + format included.
Pro	Act as marketing consultant, 10 yrs food retail. Budget: \$150k. 5 ideas + hook + format + outcome table.	Full context + role + format + outcome table.

■ You don't need fancy words — you need specificity. Every detail removes a guess the AI would otherwise make.

3.1 Common Beginner Mistakes

- Writing one vague line and hoping for the best.
- Cramming multiple unrelated tasks into a single prompt.
- Assuming the AI already knows your context.
- Accepting the first draft as final — always iterate.
- Forgetting to specify who the output is for.

4. Prompt Types Overview

Type	Description	Best For
Zero-Shot	No examples — AI uses training alone.	Quick tasks, factual queries.
Few-Shot	1–3 examples provided in the prompt.	Tone/style matching, structured outputs.
Chain-of-Thought	Ask AI to reason step by step.	Math, logic, complex decisions.
Role-Based	Assign AI a specific expert persona.	Expert-level output in any domain.
XML Structured	Use structured markup tags.	API workflows, agentic systems.
Prompt Chaining	Break task into sequential prompts.	Long docs, multi-step processes.

4.1 Key Vocabulary

Term	Meaning
Token	A chunk of text (~4 chars or ¾ of a word).
Context Window	Maximum tokens the AI can process in one session.
Temperature	Controls creativity/randomness (0 = precise, 1 = creative).
System Prompt	Background instruction given before the conversation.
Iteration	Refining a prompt based on the output received.
Hallucination	When AI produces confident but incorrect information.

■ Specificity beats cleverness. Treat your first prompt as a draft. State audience, goal, and format in every serious prompt.

5. The R-C-I-F-S Framework

The R-C-I-F-S framework is the single most important prompt structure to master. Every professional prompt should include all five elements:

Element	What It Means	Example
R — Role	Expert persona the AI should adopt.	Act as a certified financial planner with 15 years of experience.
C — Context	Background: audience, goal, constraints	Audience: first-year college students; goal: start SIP with ₹500/month.
I — Instruction	The single, clear task you want.	Create a 3-step beginner investing plan.
F — Format	How the output should look.	3-column table: Step Action Expected Outcome.
S — Structure	Logical order of the response.	Start with summary, then plan, end with one caution.

5.1 Copy-Paste Template

Role: You are a [profession] with [X] years of experience in [domain]. Context: I need this for [audience] because [goal]. Constraints: [tone / length / must-include / avoid]. Instruction: [The one clear task you want completed]. Format: [bullets / table / 150-word paragraph / numbered list]. Structure: Start with [X], then cover [Y], end with [Z].

■ Do This	■ Avoid This
Be specific about audience and goal	Write one vague line and hope
Give an example of what 'good' looks like	Cram five unrelated tasks in one prompt
Ask for one thing at a time	Assume AI knows your context
Specify length and format	Accept the first draft as final
Iterate — refine instead of restarting	Forget to say who it's for

6. Role-Based Prompting

Assigning the AI an expert persona dramatically shifts the quality and depth of responses. The richer the role definition, the more specialised the output.

Act as a [title] with [X] years of experience in [industry/domain], specialising in [specific area], working with [type of clients/audience].

Simple Role	Enhanced Role
Act as a doctor	Act as a general physician with 12 years of practice in urban India, experienced in explaining medical concepts in simple terms.
Act as a writer	Act as a senior copywriter at a D2C brand, specialising in conversion-focused landing page copy for health supplements.
Act as a teacher	Act as a Class 10 Maths teacher skilled at breaking down concepts for students with limited prior foundation.

Persona Layering Checklist

- Title — What is their job?
- Experience — How many years?
- Industry — Which sector?
- Audience they serve — Who do they work with?
- Communication style — Formal? Conversational? Technical?

■ Try the same question across 3 different expert personas and compare. The contrast shows how powerfully roles shape responses.

7. Chain-of-Thought (CoT) Prompting

CoT prompting asks the AI to reason through a problem step by step before delivering the final answer. This dramatically reduces errors on complex tasks.

7.1 Trigger Phrases

- "Think step by step"
- "Walk me through your reasoning"
- "Break this down into steps before answering"
- "Reason through this before giving a final answer"

7.2 Standard CoT Template

I need to solve: [your problem]. Think step by step: 1. List what we know 2. Break the problem into parts 3. Solve each part 4. Give a clearly labelled FINAL ANSWER

Type	How It Works
Zero-Shot CoT	Add "Think step by step" to any prompt — no examples needed.
Few-Shot CoT	Provide 1–2 worked examples with visible reasoning before the real question.

7.3 Best Use Cases

- Financial calculations and budgeting.
- Legal reasoning and case analysis.
- Technical debugging.
- Strategic decisions and trade-off comparisons.
- Project planning and scheduling.

8. Zero-Shot vs Few-Shot Prompting

Dimension	Zero-Shot	Few-Shot
Examples needed	None	1–3
Style control	Low	High
Best for	Common formats, quick tasks	Niche/style-specific outputs
When to use	Flexible quality standards, first-draft generation	Tone matching, format-sensitive outputs

8.1 Few-Shot Template

[Optional: brief task description] Example 1: Input: [sample input] Output: [desired output] Example 2: Input: [sample input] Output: [desired output] Now complete: Input: [your actual input] Output:

■ 2 examples is the sweet spot. One may not generalise; three or more can over-constrain. Always use your best work as examples.

9. XML Structured Prompting

XML prompts use structured tags to clearly separate instructions, data, context, and output requirements. Claude responds exceptionally well to XML prompting — it is the preferred format for production workflows.

Summarise the customer review below Product review for an e-commerce platform. Audience: operations team. [Paste the customer review here] - Sentiment: Positive / Negative / Neutral - Key Issue (if any): one sentence - Suggested Action: one sentence

Use Case	Why XML Helps
Document analysis pipelines	Clearly separates document from instruction.
Multi-step workflows	Each step can be tagged and referenced.
Extracting structured data	Output tags enable reliable parsing.
System prompts for products	Clean separation of persona, rules, and task.

10. Context Engineering

Model Type	Approximate Context Window
Standard models	8,000 – 32,000 tokens
Extended models	100,000 – 200,000 tokens
Claude 3+	Up to 200,000 tokens

10.1 Memory Block Technique

When a conversation grows long, summarise key context into a compact block and re-paste it at the start of a new session:

MEMORY BLOCK (paste at start of each new session): - Project: Redesign of loyalty program - My role: UX Consultant - Client constraints: Budget under Rs.10L, launch by March - Decisions made: Tier system confirmed (Silver/Gold/Platinum) - Last task completed: User journey map draft - Next task: Wireframe for onboarding flow

10.2 Fresh vs Continue Decision

Situation	Recommendation
New unrelated task	Start fresh conversation.
Continuing same project	Keep thread; add memory block if long.
AI seems confused or off-track	Fresh session with clean compressed summary.
Output quality degrading	Fresh session with better initial prompt.

11. Prompt Chaining

Break a complex task into a sequence of smaller prompts where each output feeds into the next. This mirrors a professional workflow and produces far higher quality on tasks too complex for a single prompt.

Prompt 1 → Output 1 | Prompt 2 (uses Output 1) → Output 2 | Prompt 3 (uses Output 2) → Final Output

Step	Prompt
1 — Research	Act as a content strategist. List 7 data-backed reasons why sleep deprivation harms productivity. One statistic per point.
2 — Outline	Using these 7 points [paste Output 1], create a structured outline for a 1,000-word blog post targeting stressed corporate professionals.
3 — Draft	Using this outline [paste Output 2], write the full 1,000-word blog post. Tone: conversational and empathetic.
4 — Polish	Review this blog post [paste Output 3]: 1. Improve headline 2. Tighten paragraphs 3. Strengthen CTA.

11.1 Use Cases

- Long-form content (books, reports, courses).
- Research → analysis → recommendation workflows.
- Job application: research → tailor CV → write cover letter.
- Product development: ideation → user stories → feature specs.

12. AI Agents & Agentic Prompting

Concept	Description
Tool use	AI can call external functions (web search, code execution, APIs).
Memory	Persistent context across sessions (short-term or long-term).
Planning	AI breaks a goal into sub-tasks autonomously.
Action loops	AI takes action, observes result, takes next action.
Human-in-the-loop	Points where human review or approval is required.

12.1 System Prompt Template for Agents

You are a research assistant agent. Goal: answer questions by searching, synthesising, and citing sources. TOOLS: web_search, document_reader, calculator RULES: - Verify claims using at least 2 sources - Ask for clarification before irreversible actions - Summarise your plan before executing - Report errors immediately OUTPUT: 1. Action taken 2. Finding 3. Source 4. Confidence: High/Medium/Low

12.2 Safety Principles

- Minimal footprint: Request only necessary permissions.
- Confirm before irreversible actions: file deletion, sending emails, payments.
- Pause points: Build in check-ins at key decision points.
- Clear error handling: Tell the agent what to do with unexpected input.

13. Claude-Specific Best Practices

Tip	How to Apply
Use XML tags	Claude responds exceptionally well to XML-structured prompts.
Be explicit about thinking	Use: "Think carefully before answering" or "Work through this step by step."
Leverage system prompt	Put persona, rules, constraints in system prompt via API — keep user messages clean.
Long-context strength	Feed entire contracts, reports, or transcripts for deep analysis.

Tip	How to Apply
Self-critique loop	"Review your answer. What is the weakest part? How would you improve it?"

13.1 Claude System Prompt Template (API)

```
# SYSTEM PROMPT You are [role] with [experience]. Your communication style is [tone].  
Always [key rule 1] and [key rule 2]. Never [key restriction]. # USER MESSAGE [Clear single  
instruction] [Background, audience, goal] [User content or question] [Exact format  
required]
```

14. Advanced Prompting Techniques

Technique	When to Use	Key Phrase / Template
Persona Contrast	Business decisions needing multiple angles	Answer from 3 perspectives: CFO, VC, bootstrapped founder.
Socratic Prompt	Learning or strategy before jumping to answer	Ask me the 5 most important questions you need answered."
Reverse Prompting	Unsure how to prompt for your task	"What would the ideal prompt for this task look like?"
Devil's Advocate	Before committing to a decision	"Give me the 5 strongest arguments AGAINST this decision."
Constraint Creativity	Creative tasks (copy, taglines, concepts)	Exactly 5 words. Must include a verb. Cannot use: [words].
Self-Critique Loop	Improving any AI output	"Review your answer. What is the weakest part? Rewrite it."
Meta-Prompting	Diverse tasks with one universal prompt	Design a single prompt reusable for diverse tasks via additional instructions
Least-to-Most	Complex problems requiring specialisation	Start general, then add facts to drive a specialised solution.

15. Quick-Reference Cheat Sheet

Goal	Technique to Use
Match a specific style	Few-Shot
Complex reasoning	Chain-of-Thought
Expert-level output	Role-Based
Long complex project	Prompt Chaining
API / developer use	XML Structured
Long documents	Context Engineering
Multiple perspectives	Persona Contrast
Idea generation	Constraint-Based Creativity
Improve AI output	Self-Critique Prompting

15.1 The 60-Second Prompt Formula

```
"Act as a [ROLE] with [EXPERIENCE]. For [AUDIENCE], [TASK VERB] [OBJECT]. Context: [GOAL +  
CONSTRAINT]. Format: [OUTPUT FORMAT]. Length: [WORD COUNT]."
```

15.2 Prompt Quality Checklist

- Is there a clear role assigned?
- Is the audience specified?
- Is the goal / purpose stated?
- Is the task a single clear instruction?
- Is the format specified (length, structure)?
- Are constraints mentioned?
- Have I provided an example if needed?
- Am I prepared to iterate on the first output?

15.3 Prompt Failure Diagnosis

Problem	Likely Cause	Fix
Too generic	No role or audience	Add role + audience context
Wrong format	No format instruction	Specify bullets/table/word count
Too long	No length constraint	Add 'under X words'
Off-topic	Task too vague	Use one clear verb + object
Missing my style	No examples	Add 1–2 few-shot examples
Incorrect reasoning	No step-by-step	Add 'think step by step'
Loses thread	Context window issue	Add memory block

16. Prompt Engineering Techniques (GFG)

16.1 Foundational Techniques

Technique	Description
Information Retrieval	Create prompts so the LLM retrieves relevant knowledge from its training base.
Context Amplification	Add supplementary context to direct the LLM's understanding and focus.
Summarisation	Induce the LLM to generalise or write summaries about complex themes.
Reframing	Rephrase prompts to make the LLM consider a specific style or format.
Iterative Prompting	Break complex tasks into smaller parts and instruct the LLM sequentially.

16.2 Advanced Techniques

Technique	Description
Least to Most	Start with general prompts, add specificity to drive highly specialised solutions.
Chain-of-Thought	Require the LLM to show reasoning steps alongside the answer.
Self-Ask Prompting	Prompt the LLM to ask itself clarifying questions (variant of CoT).
Meta-Prompting	Design one universal prompt reusable for diverse tasks via additional instructions.

16.3 Best Practices (GFG)

- Begin with Objectives and Goals — know your purpose before crafting the prompt.
- Use Relevant and Specific Data — avoid irrelevant training data.
- Focus on Relevant Keywords — 'mathematics' ensures 'planes' is defined correctly.
- Keep Prompts Simple and Clear — plain language avoids misinterpretation.
- Test and Refine — evaluate with different test cases and continuously improve.

16.4 Advantages & Disadvantages

Advantages	Disadvantages
Improved accuracy — relevant prompts produce refined, precise responses.	Difficulty in determining specificity — too specific limits range; too general produces irrelevant results.
Enhanced user experience — users get relevant responses without extra hassle.	
Cost-effective — fewer rounds needed to reach a satisfactory response.	

16.5 Future of Prompt Engineering

- AI and NLP — prompts will become more accurate and use-case specific as models advance.
- Integration with Other Technologies — virtual assistants, chatbots, and voice devices.
- Increased Automation — more advanced prompts will streamline the prompt development process itself.

17. 30-Day Practice Plan

Week 1: Foundations

Day	Task
1	Write 5 prompts using different styles. Notice what changes.
2	Use the full R-C-I-F-S skeleton on a real work task.
3	Try "think step by step" on a complex decision you face.
4	Ask the same question as 3 different expert personas.
5	Give 2 examples (few-shot) and compare to zero-shot output.
6	Take one output and refine it with 3 follow-up nudges.
7	Save your 3 best prompts as reusable templates.

Week 2: Intermediate

Day	Task
8	Write an XML-structured prompt for a complex task.
9	Build a 3-prompt chain for a blog post or report.
10	Use the Socratic prompt technique for a topic you want to learn.
11	Use persona contrast on a business decision.
12	Use the self-critique technique on your best recent AI output.

Day	Task
13	Summarise a long document into a compact memory block.
14	Review your prompt library. Improve 3 templates.

Week 3: Application

Day	Task
15	Write 5 prompts for your most common work tasks.
16	Use constraint-based creativity for a marketing task.
17	Use reverse prompting — ask AI what a great prompt looks like.
18	Build a prompt for a business use case (sales, ops, HR).
19	Use the devil's advocate technique on a real decision.
20	Practice a few-shot data extraction prompt.
21	Full project using chaining: research → outline → draft → edit.

Week 4: Mastery

Day	Task
22	Build a Claude system prompt for a persona you use regularly.
23	Design a simple agentic prompt with tool use specifications.
24	Create a prompt template library with 20 entries.
25	Run the interview Q&A from memory.
26	Teach someone else one technique (teaching = mastery test).
27	Find a real business problem and solve end-to-end with prompting.
28	Write your personal "Prompting Philosophy" — 5 principles you follow.
29	Identify your top 3 weak areas. Practice one prompt in each.
30	Publish your learnings — LinkedIn post, Notion doc, or portfolio piece.

18. Interview Preparation Q&A;

Q1: What is prompt engineering?

The practice of crafting instructions (prompts) that guide AI models to produce accurate, relevant, and useful outputs. The same AI can deliver vastly different results depending on how it is instructed.

Q2: Explain the R-C-I-F-S framework.

R-C-I-F-S = Role, Context, Instruction, Format, Structure. Role sets the AI's expert persona; Context gives background and constraints; Instruction states the single clear task; Format specifies output presentation; Structure defines the logical order.

Q3: What is CoT prompting and when do you use it?

CoT asks the AI to reason step by step before giving a final answer. Most useful for complex multi-step problems — math, logical reasoning, troubleshooting, and decision comparisons — where jumping to a conclusion increases error risk.

Q4: Difference between zero-shot and few-shot prompting?

Zero-shot gives the AI no examples — it relies on training alone. Few-shot provides 1–3 input-output examples so the AI learns the exact style required. Few-shot is more reliable for style-specific, format-sensitive, or niche outputs.

Q5: What is a context window and how does it affect prompting?

The context window is the maximum text an AI can process in one interaction. As conversations grow, earlier content receives less attention. Good context engineering means summarising earlier content into memory blocks and knowing when to start a fresh session.

Q6: What is prompt chaining?

Breaking a complex task into a sequence of smaller prompts, where each output feeds into the next. Enables higher quality on tasks too complex for a single prompt, mirroring a professional workflow: research, outline, draft, edit.

Q7: What are XML prompts and why are they used in production?

XML prompts use structured tags to separate instructions, data, context, and output requirements. Preferred in production because they reduce ambiguity, enable reliable output parsing, and keep complex multi-part instructions organised.

Q8: How do you handle a prompt that keeps producing poor results?

Use Prompt Failure Diagnosis: identify whether the issue is vagueness (add role/audience), format (specify explicitly), length (add word count), reasoning (add step-by-step), or style (add few-shot examples). Iterate with targeted refinements.

Q9: What is agentic prompting?

Writing instructions for AI systems that take autonomous action sequences using tools, memory, and planning. Includes specifying available tools, behaviour rules, decision checkpoints, and error-handling protocols.

Q10: How would you build a prompt library for a business?

Identify the 10–15 most repeated tasks. Write and test an R-C-I-F-S prompt for each. Save as templates with [BLANK] placeholders. Organise by department or use case. Assign ownership, version control, and a review cycle.

Remember: Your first prompt is a draft, not a verdict. Professionals don't nail it in one shot — they nudge, correct, and refine. Two or three follow-ups usually beat one 'perfect' prompt.

Sources: GeeksforGeeks — What is Prompt Engineering (2025) | ABTalks Prompt Engineering Study Handbook (Edition 2025)